Bett's law

$$\sum P_{iA} V_{iB} = \sum P_{iB} V_{iA}$$

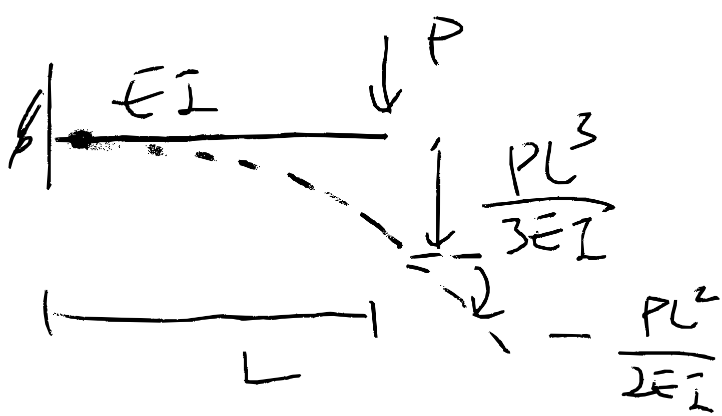
1 & 2 $\quad 10 \times 1.59 + 15 \times 1.82 = 20 \cdot V_c$

1 & 3

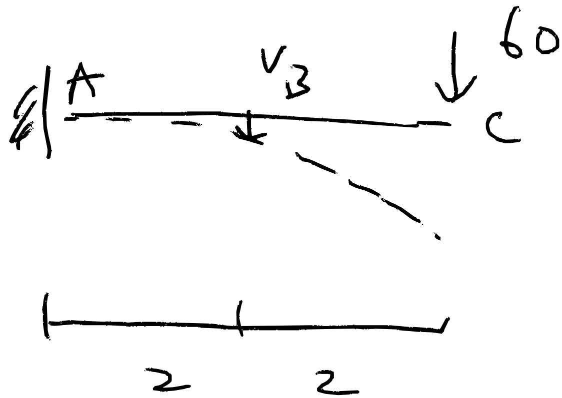
$$10 \times 1.52 + 15 \times 1.65 = 20 \cdot V_c + 5 \times (-0.67)$$

Maxwell Example 1

Assume that we have the following

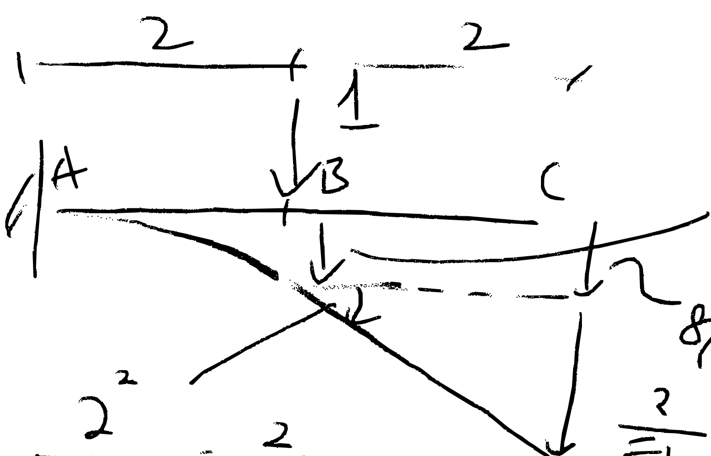


Example 1



$$V_B = f_{BC} \cdot 60$$

According to Maxwell's Reciprocal theorem : $f_{BC} = f_{CB}$



$$\frac{2^3}{3EI} = \frac{8}{3EI}$$

$8/3EI$

$$\frac{2^2}{2EI} = \frac{2}{EI}$$

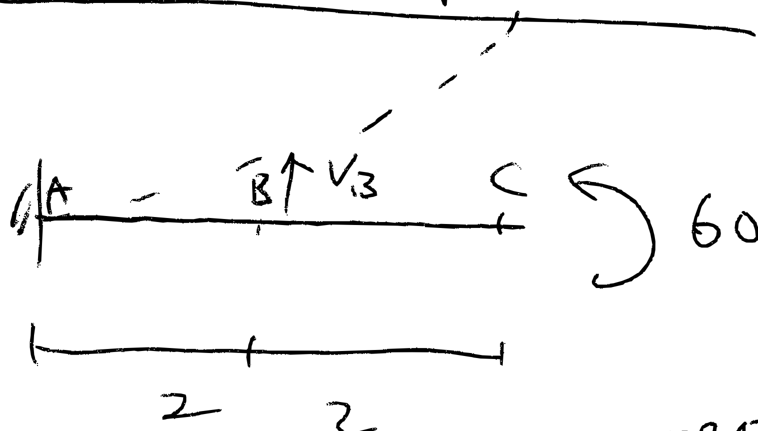
$$\frac{2}{EI} \times 2 = \frac{4}{EI}$$

$$f_{CB} = \frac{8}{3EI} + \frac{4}{EI} = \frac{20}{3EI}$$

$$V_B = 60 f_{CB}$$

$$= 60 \times \frac{20}{3EI} = \frac{400}{EI} \quad (\downarrow)$$

Maxwell Example 2

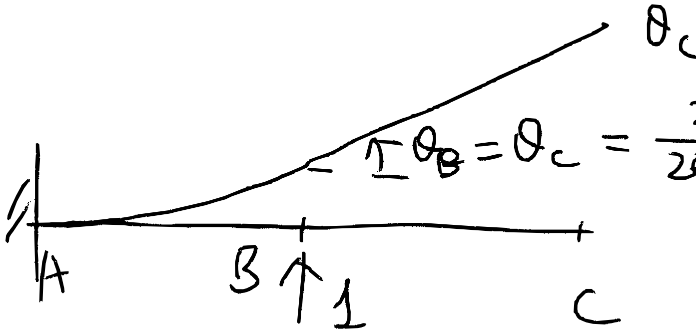


$$V_B = \underset{\text{for moment}}{f_{BC}} \cdot 60$$

$$f_{BC} = f_{CB} \quad \text{(per Maxwell)}$$

rotation

unit load at B

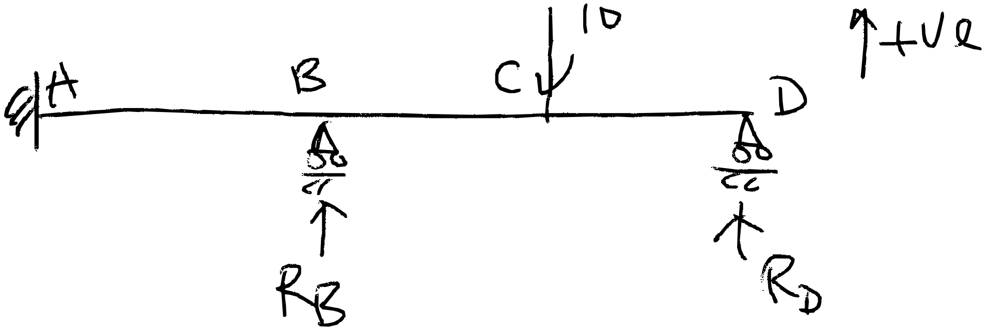


$$\theta_B = \theta_C = \frac{2^2}{2EI} = \frac{2}{EI}$$

$$f_{CB} = \frac{2}{EI}$$

$$V_B = f_{CB} \cdot 60 = \frac{120}{EI}$$

Maxwell Example 3



Two redundants : R_B & R_D

Compatibility conditions : $V_B = 0$

$V_D = 0$

$$V_B = \underbrace{V_{B1}}_{\text{Pt. load } 10} + \underbrace{V_{B2}}_{R_B} + \underbrace{V_{B3}}_{R_D} = 0$$

$$V_{B1} = f_{BC}(-10)$$
$$V_{B2} = f_{BB} R_B$$
$$V_{B3} = f_{BD} R_D$$

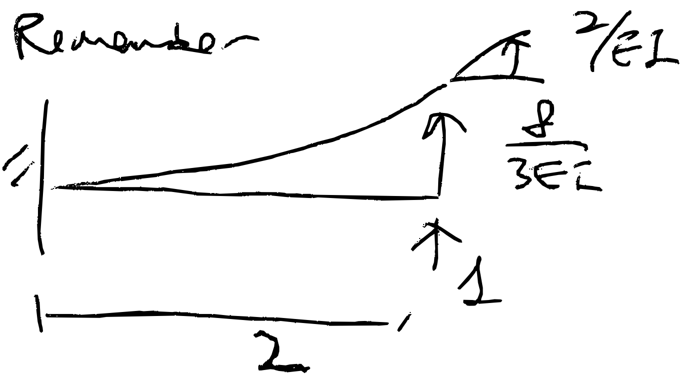
$$0 = -10 f_{BC} + f_{BB} R_B + f_{BD} R_D$$

$$V_D = \underset{\substack{\uparrow \\ (-10)}}{V_{D1}} + \underset{\substack{\uparrow \\ R_B}}{V_{D2}} + \underset{\substack{\uparrow \\ R_D}}{V_{D3}} = 0$$

$$V_{D1} = f_{DC}(-10)$$
$$V_{D2} = f_{DB} R_B$$
$$V_{D3} = f_{DD} R_D$$

$$0 = -10 f_{DC} + f_{DB} R_B + f_{DD} R_D$$

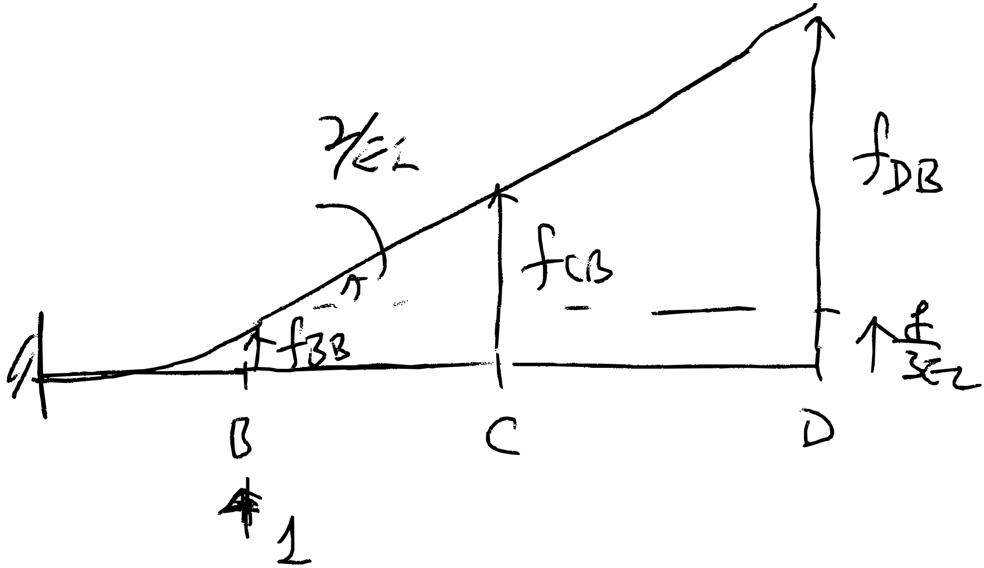
Remember



$$f_{BB} = \frac{8}{3EI}$$

$$f_{BD} = f_{DB} =$$

$$f_{BC} = f_{CB}$$



$$f_{BB} = \frac{8}{3EI}$$

$$f_{CB} = \frac{8}{3EI} + \frac{2}{EI} \times 2 = \frac{20}{3EI}$$

$$f_{DB} = \frac{8}{3EI} + \frac{2}{EI} \times 4 = \frac{32}{3EI}$$

Compatibility

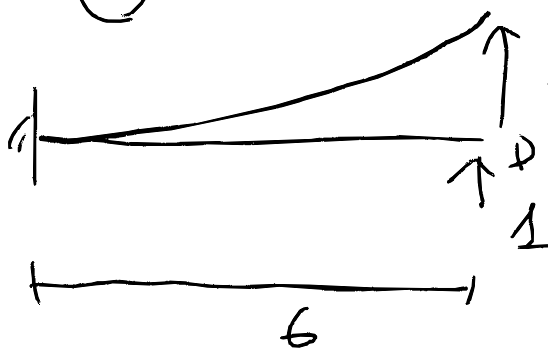
$$0 = -10 f_{BC} + f_{BB} R_B + f_{BD} R_D$$

$$= -10 f_{CB} + f_{BB} R_B + f_{DB} R_D \quad \text{— (1)}$$

2^{nd} : $v_c = 0$

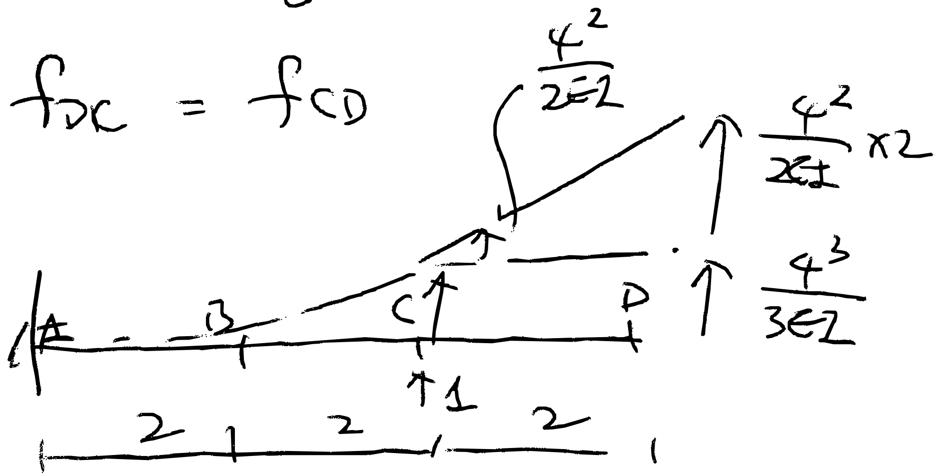
$$0 = -10 f_{DC} + f_{DB} R_B + f_{DD} R_D$$

$- ②$ ✓ ✓ ↑



$$f_{DD} = \frac{6^3}{3EI} = \frac{72}{EI}$$

$$f_{DC} = f_{CD}$$



$$f_{DC} = \frac{4^3}{3EI} + \frac{4^2}{2EI} \times 2$$

$$= \frac{112}{3EI}$$

In matrix form for ① & ②

$$\begin{bmatrix} f_{BB} & f_{BD} \\ f_{DB} & f_{DD} \end{bmatrix} \begin{bmatrix} R_B \\ R_D \end{bmatrix} = 10 \begin{bmatrix} f_{BC} \\ f_{DC} \end{bmatrix}$$

$$f_{BB} = \frac{8}{3EI}$$

$$f_{BD} = f_{DB} = \frac{32}{3EI}$$

$$f_{DD} = \frac{72}{EI}$$

$$f_{BC} = \frac{20}{3EI}$$

$$f_{DC} = \frac{112}{3EI}$$

Once solved,

$R_B = 10.45$ (↑)
$R_D = 3.64$ (↑)